10.1 Measuring an angle in radians

Key point · Definition and results. The unit circle is the circle with centre O and radius 1, oriented anticlockwise. One radian corresponds to an arc of length 1 on this circle. A full turn is 2π radians, a half turn π. On a circle of radius r, an arc length is $r\theta$ for a positive angle θ expressed in radians.

Status and justification. Definitions and proportionality between central angle and arc length. The conversion is $180° = \pi$ radians; we multiply degrees by $\frac{\pi}{180}$.

Example 1. $60° = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270°$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of 2π cm.

Watch out. The length of an arc is not that of the chord. In trigonometric calculations, check the degree or radian mode of the tool.

Application. Convert 45°, 120°; give the arc for radius 2 m and angle $\frac{3\pi}{4}$.

10.2 Wrapping and the image of a real number

Key point · Definition and results. To each real number x corresponds a point M of the circle, obtained by an oriented path of length x starting from I(1;0). Two real numbers have the same image if their difference is an integer multiple of 2π. A negative number indicates a path in the clockwise direction.

Status and justification. Adding or removing a whole number of turns keeps the end point. Conversely, returning to the same point requires a whole number of turns.

Example 2. $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: the point $(0, -1)$.

Watch out. Adding π gives the opposite point, not the same point.

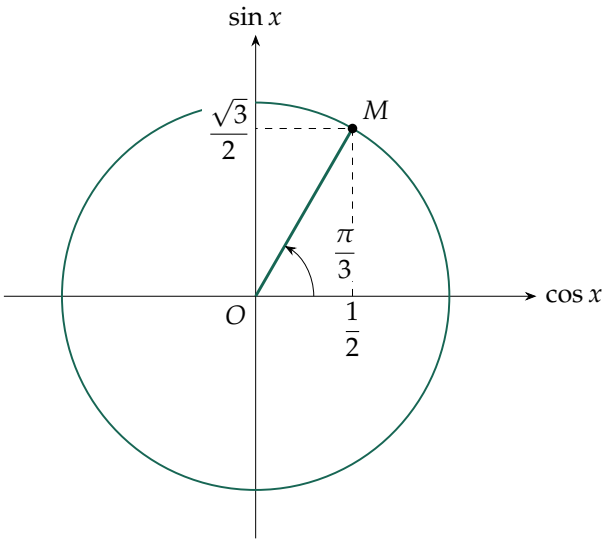
Application. Place the images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ by reducing them to within one turn.

10.3 Sine, cosine and standard values

Key point · Definition and results. If M is the image of x, $M = (\cos x, \sin x)$. So $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$. For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosines are $1, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, 0$ and the sines $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Status and justification. At $\frac{\pi}{4}$, the two coordinates are equal and positive; Pythagoras gives $2a^2 = 1$, so $a = \frac{\sqrt{2}}{2}$. An equilateral triangle of side 1, cut by its altitude, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. The complementary angle gives $\frac{\pi}{6}$.

> **Example 3.** The point at angle $\frac{\pi}{3}$ is $(\frac{1}{2}, \frac{\sqrt{3}}{2})$. The scalar product of two unit vectors forming this angle equals its cosine, so $\frac{1}{2}$. The geometry chapters thus connect to the circle.



For the angle $\frac{\pi}{3}$, the projections of M give its cosine and sine directly. The fractions are placed outside the lines of the drawing.

> **Watch out.** The cosine is the x-coordinate, the sine the y-coordinate. The identity gives an absolute value: the quadrant determines the sign.

> **Application.** If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

10.4 Related angles and symmetries

> **Key point · Definition and results.** The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Adding $2k\pi$, k an integer, changes neither sine nor cosine.

Status and justification. Each equality can be read from the coordinates of the symmetric points: horizontal axis for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a catalogue without justification.

Example 4. $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry from $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Watch out. Two angles with the same cosine do not necessarily have the same sine. Reduce the angle, then read the signs.

Application. Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

10.5 Approximating pi and putting it to use

Key point · Definition and results. A regular inscribed polygon with n sides in the unit circle has half-perimeter $n\sin(\frac{\pi}{n})$; a circumscribed polygon has half-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ here. Archimedes' method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi in advance.

Status and justification. The inscribed perimeter is less than that of the circle and the circumscribed one greater: geometric bounds accepted. If c is the chord of one side, let $h = \sqrt{1 - \frac{c^2}{4}}$; the chord of the half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids a subtraction of nearby real numbers.

Example 5. For the inscribed hexagon, c=1 and half-perimeter 3. The circumscribed one gives $2\sqrt{3} \approx 3.4641$. By doubling the sides, we tighten the bounds. The supplied program starts from n=6 and c=1.

Watch out. Using a sine function directly with pi already known would check a calculation, but would not construct an approximation independent of pi.

Application. For n=6 and c=1, calculate the new chord after doubling.